

CHAPTER 3

Question 3 **(15 marks)**

In January of 2006, Adam family moved to a tropical climate country. For the following year (2007), Adam has recorded the number of rainy days that occurred each month. The data from January to December contained 14, 14, 10, 12, 11, 13, 11, 11, 14, 10, 13, 8.

a. Find the mean, mode, median and range for the data set of rainy days. [5 marks]

b. What can you tell about the distribution of the data based on the median and mean found in (a)? [2 marks]

c. Find the standard deviation for the data set of rainy days above. [2 marks]

d. Suppose the number of rainy day in December has been recorded wrongly. The observed data is 7 not 8. Does this data can be considered an outlier? Support your answer.

e. In year 2008, the number of rainy days has been recorded and the standard deviation is 2.5. Based on the standard deviation found in (c),in which year does the number of rainy days show a more dispersed distribution? Justify your answer. [2 marks]

Question 3

(10 MARKS)

Data on the scores for 31 students taking a mathematic test were collected and presented as histogram in Figure 1 below.

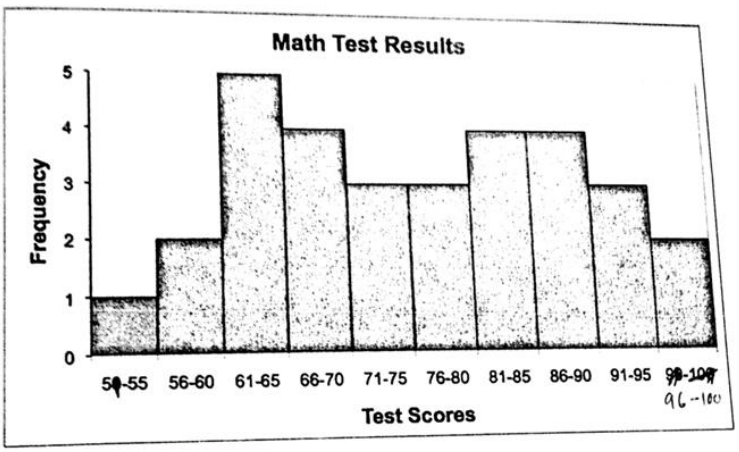


Figure 1: Histogram of Student's Mathematic Test Results

(a) State the mode class of the student's test scores. [1M]

(b) Find the mean of the student's test scores. [4M]

(c) Find the median of the student's test scores. [4M]

(d) Determine the skewness of the graph obtained from the histogram. [1M]

Mid-Term 2016/2017

Thirty students enrolled in "Statistic Using SPSS" course. The course materials, grades and lecture notes are available to students on e-learning website where frequency of access of each student to any of the pages were recorded. Table 3 below shows frequency of access for the thirty students after one month period.

Class Interval	Frequencies
0 – 4	1
5 – 9	5
10 – 14	5
15 – 19	6
20 – 24	5
25 – 29	4
30 – 34	2
35 - 39	1
40 - 44	1

Find median of the number of access. (4 marks)

Find mode of the number of access. (4 marks)

Mid-Term 2017/2018

QUESTION 6 **(10 MARKS)**

Table 5 give data on the heights, in cm, of 51 children

Table 5

Class interval	$140 \le h \le 150$	$150 \le h \le 160$	$160 \le h \le 170$	$170 \le h \le 180$
Frequency	6	16	21	8

a) Estimate the mean height [3 marks]

b) Estimate the median height [3 marks]

c) Find the modal class [4 marks]

Mid-Term 2018/2019

Question 3 **[15 MARKS]**

(a) Akmal designed a game where customers pay RM2 and are given 14 balls which they have to try to get into a container which is 6 metres away. If they get more than 8 balls in the container, they win RM5. Before introducing the game, she tries it out on a sample of 50 people. The number of successes scored by each person is recorded and grouped as below.

Table 3

Score	**Frequency**
0 – 2	9
3 – 5	15
6 – 8	20
9 – 11	4
12 – 14	2

Calculate

(i) mode (2 marks)

(ii) median (2 marks)

(iii) mean (2 marks)

(b) As a quality controller, Eric collected data relating to the life expectancy of a sample of 60 light bulbs produced by company XYZ. The mean life was 650 hours and the standard deviation was 8 hours. A second sample of 80 bulbs was taken by Allen and resulted in mean life of 660 hours and standard deviation 7 hours. Find the **overall mean** for the light bulbs life expectancy produced by the company. (2 marks)

(c) A civil engineer monitors water quality by measuring the amount of suspended solids in a sample of river water. Over 5 days, she observed suspended solids (parts per million) as below:

14 12 21 28 30

(i) Given the mean sample as 21 parts per million, find the variance and standard deviation. (3 marks)

(ii) Find the skewness value for the sample. (2 marks)

(iii) From your answer in (ii), is this sample positively skewed or negatively skewed? Justify your answer. (2 marks)